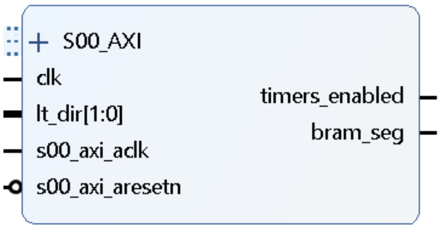
Timers

1. Introduction

The Timers IP core allows to control how the acquisition is running, and to monitor real and live time. The acquisition can run either in Manual or Auto mode. In the manual mode the acquisition stops when real time or live time reaches user defined value. In the Auto mode, the acquisition is automatically restarted each time the preset is reached and acquiring of the data are done in two memory segments: each time new run is started, the segment is switched. User can access both segments during measurement. It is upon user to read and clear memory segment that is not currently in use. There is no extra dead time introduced when measurement is restarted automatically.

2. Usage



The IP core has five internal registers which allow user to tailor the measurement (see table below) and monitor live and real timer. The registers can be accessed via standard AXI bus interface S00_AXI at offset address 0. The IP core will be mapped in a processor's memory address space and it will get a fixed base address. The input port *lt_dir* is used to control live time counter and it should be connected to the same name output port of the ip_pur IP core. The output port *bram_seg* is used to define in which memory segment data are stored. In the Auto mode it toggles, that is changes value between 1 and 0, each time a new measurement is started. In the Manual mode it keeps 0 value. The port should be connected to corresponding port of *ip_bram_incr* and *ip_roi_counter* . The Manual and Auto mode can be selected with using status register 5, bits 0 and 1.

The Real time and Live time can be measured by any of the three timer counters named A, B and C. Only counter C can have preset. The counters run using system clock (50 Mhz). However, the clock is enabled with using another counter. It can be either an output of a free running counter, or a counter which is controlled by lt_dir port. In both cases the enable counters are effectively measure 50 000 system clocks. This means the following: live timer is incremented every 1 msec; live timer is incremented after system is live for interval of 1 msec; the precision of the real and live time measurement is 1 msec.

The timers are controlled by status register 5: their function (real timer or live timer) can be assigned with bits 2, 3 and 4; they can be enabled or disabled with bits 5, 6 and 7; they can be cleared with bits 8, 9 and 10. The timers which are enabled, starts counting when status register 5, bits 0 (Manual mode) or 1 (Auto mod) are set to 1.

In the Manual mode the counters stop when the bit 0 is set to 0. The bit can be set to 0 either by user (to stop counting manually) or by ip_core when counter C reach preset (in this case bit 0 will remain 1).The running status of the counter C can be obtained by reading register 0, bits 0 or 1. Note that bit 0 can be changed only by user, therefore it will stay 1 after C stops. It means that user should use bit 1 to check if C is stopped and then clear bit 0 before starting new measurement.

In the Auto mode, the counters never stop counting. When the counter C reaches preset, all enabled counters are cleared and restarted. In order to find if the counters were restarted in the Auto mode, you can read register 3, bit 2 twice and check if its value has toggled.

<table>
<tr><th>Name</th><th>Meaning</th><th>Description</th></tr>
<tr><td>S00_AXI</td><td>Port</td><td>AXI bus signals</td></tr>
<tr><td>s00_axi_aclk</td><td>Port</td><td>AXI bus clock</td></tr>
<tr><td>s00_axi_arestn</td><td>Port</td><td>AXI bus reset</td></tr>
<tr><td>lt_dir[1:0]</td><td>Port</td><td>Output port which controls counting of the Live timer</td></tr>
<tr><td>bram_seg</td><td>Port</td><td>Output port. In Manual mode it keeps value 0. In Auto mode it toggles between 1 and 0</td></tr>
<tr><td>clk</td><td>Port</td><td>Input clk, must be the same clock which is used for data ports</td></tr>
<tr><td>C S00_TIMERS_DATA_WIDTH</td><td>Instantiation Parameter</td><td>maximal length of the delay $L = 2^{C\_ADDRESS\_WIDTH}$ (10 is default)</td></tr>
<tr><td>slv_reg0</td><td>Register</td><td>32 bit, read-only register, which contains number of counts counted by Counter A. Read this register at address 0x0 to get current number of the counts.</td></tr>
<tr><td>slv_reg1</td><td>register</td><td>32 bit, read-only register, which contains number of counts counted by Counter B. Read this register at address 0x4 to get current number of the counts.</td></tr>
<tr><td>slv_reg2</td><td>register</td><td>32 bit, read-only register, which contains number of counts counted by Counter C. Read this register at address 0x8 to get current number of the counts.</td></tr>
<tr><td>slv_reg3</td><td>register</td><td>32-bit, read-only register which defines running status of the counters as described below. Read this register at address 0xC in order to obtain the status.
<table>
<tr><th>bit</th><th>function</th></tr>
<tr><td>0</td><td>Set to 1 if Counter C did not reach preset and one of the following bits are set to 1: slv_reg5[0] or slv_reg5[1]</td></tr>
<tr><td>1</td><td>Set to 1 if Counter C reached preset, otherwise set to 0</td></tr>
<tr><td>2</td><td>Set to index of the current active bram segment (either 0 or 1)</td></tr>
<tr><td>3-31</td><td>Not used</td></tr>
</table></td></tr>
<tr><td>Slv_reg4</td><td>Register</td><td>32 bit, read-write register, which defines preset for the counter C. Write to address 0xF to set the preset. Read from the same address to get current preset value.</td></tr>
<tr><td>Slv_reg5</td><td>register</td><td>32 bit, read-write register which defines status of the timers as described below. Write to address 0x14 to set status. Read from the same address to get current status. If you want to change only one bit,</td></tr>
</table>

		then first read all bits, change the bit and finally write back.	
		Bit	function
		0	Manual mode: all enabled counter starts counting when this bit is set to 1 and they stop when preset in counter C is reached. Upon preset reg3-bit0 is also set. Write to this register to set Manual mode.
		1	Auto mode: all enabled counter starts counting when this bit is set to 1 and they stop when preset in counter C is reached. After that all counters are automatically cleared, reg3-bit1 is toggled and new counting cycle is started. The number of cycles is not limited. Write to this register to set Auto mode.
		2	counter C counts live time if this bit is 1, otherwise it counts real time
		3	counter B counts live time if this bit is 1, otherwise it counts real time
		4	counter A 1rcounts live time if this bit is 1, otherwise it counts real time
		5	counter C is enabled if this bit is set to 1
		6	counter B is enabled if this bit is set to 1
		7	counter A is enabled if this bit is set to 1
		8	counter C is cleared if this bit is set to 1
		9	counter B is cleared if this bit is set to 1
		10	counter A is cleared if this bit is set to 1
		11-31	Not used

3. Architecture

The ip core comprises standard AXI bus VHDL modules generated by the template and a custom written timers.vhd module. The standard AXI modules a slightly modified in order to add custom ports and instantiate timers module.

The timers.vhd module features three equal 32-bit Master counters (MC) A, B and C (MCA, MCB, MCC) which counts using 50 MHz clock, in the ADC clock domain. They also have clock enable and clear inputs. The only output port is 32-bit number of counts. The output of the MCC feeds one input of a comparator and comparator's output is used to construct MCC's clock enable input. This preset register (AXI bus register 4) feeds the second input of the comparator.

The clock enable input of each MC is controlled by one Clock-Enable Counter (CEC). There are two types of CEC: Free Running CEC (FR-CEC) and Live Time CEC (LT-CEC). Number of FR-CEC is three and number of LT-CEC is one.

Each FR-CEC is 16-bit, runs on the 50 MHz clock of the ADC domain, and restarts counting when counts out 49999. It means that output of a FR-CEC, when connected to corresponding MC, will increment the MC every millisecond. In other words, they will measure time in miliseconds

The LT-CEC also runs on the the 50 MHz clock of the ADC domain, however it is controlled by the lt_dir[1:0] input port. The lt_dir[1:0] signals should be encoded according to Gedcke-Hale method as three-fold state: count forward, count backward and hold (stop counting), see also documentation for PUR IP core. We also want that a MC which is connected to LT-CEC measures cumulative live time in milliseconds. The LT-CEC can be implemented on the similar way like the FR_CEC: to make it 16-bit, turn around it at 49999, and count according to the Gedcke-Hale counter (see table below). In order to avoid counting down (and problems with multiple turns around during once cycle), we increased number of LT-CEC bits to 17, turn it around at 99999 and used counting logic as shown in the table below. It is obvious that such LTE-CEC counts two times faster in comparison to corresponding Gedcke-Hale LTE-CEC. However, it also turns around at two times larger number of counts, therefore bouth will trigger the MC at the same time, which means at the time when cumulative live time reaches 1 millisecond.

Lt_dir[1:0]	Gedcke-Halle	This work
00	Increment for 1	Increment for 2
01	Decrement for 1	Hold
10	Hold	Increment for 1

There are three multiplexers which route clock enable signal to MCA, MCB, and MCC. They are controlled by register 5, bits 2, 3 and 4, respectively.

All counters are enabled and cleared by two type of sources. The static sources change only on the intervention of the user and dynamic changes by the system. The static sources are counter enable bits 5, 6 and 7 and counter clear bits 8, 9 and 10 of the AXI register 5. Also, manual start bit0 of the register 5 is used to disable counters. The dynamic source which can stop counters is counter C terminate signal from comparator.

In the Auto mode there are additional enable and clear sources. These are auto_en and auto_clr signals that can stop and clear counters. They are generated by the fsm_auto_mode finit state machine (FSM). The FSM has three state: IDLE, CLEAR and COUNT. In the IDLE state the FSM is not active. When user sets Auto mode flag (by setting register 5, bit 1, to 1) the machine goes to CLEAR state for one clock and then makes transition to COUNT state. In the CLEAR state, the FSM clears all counters. In the COUNT state it waits for terminal count signal from comparator and then returns back to CLEAR state. The cycle terminates only when user clears Auto mode flag (sets register 5, bit 1 to 0)